DID $\quad y_{it} = \alpha + \beta D_i + \delta t_t + \tau D \times t_{it} + \varepsilon_{it}$

Primeras-Diferencias = DID

$$Y_{it} = \alpha + \beta D_i + \delta t + \underline{\tau} D \times t + \varepsilon_{it} .$$

$$\Delta Y_i = Y_{it} - y_{it-1}$$

$$Y_{i,t=1} = \cancel{\alpha} + \cancel{\beta D_i} + \delta + \tau D + \varepsilon_{it=1}$$

$$Y_{i,t=0} = \cancel{\alpha} + \cancel{\beta D_i} + \varepsilon_{it=0}$$

$$\Delta y_i = Y_{i,t=1} - y_{i,t=0}$$

$$\rightarrow \Delta y_i = \delta + \underline{\tau} D + \Delta \varepsilon_i$$

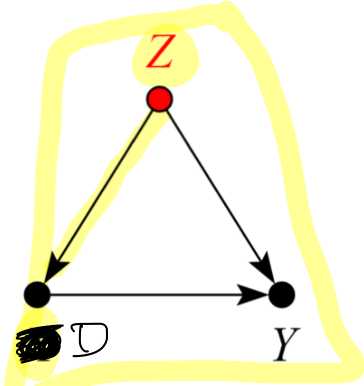
reshape

i	$Y_{1,t=0}$	$Y_{it=1}$	D_i	$X_{it=0}$	$X_{it=1}$
1					
2					
3					
4					
5					
1					

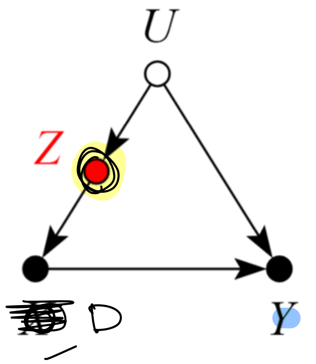
$$\Delta Y_i = \delta + \tau D + X\beta + \Delta\varepsilon_i$$

$Z_1 \ldots$

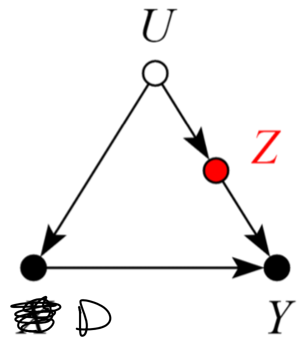
Buen Control.



(1)

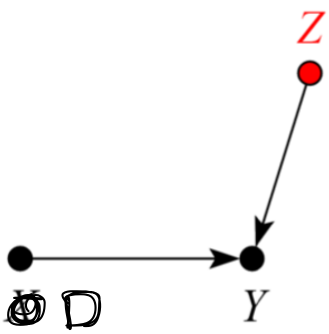


(2)



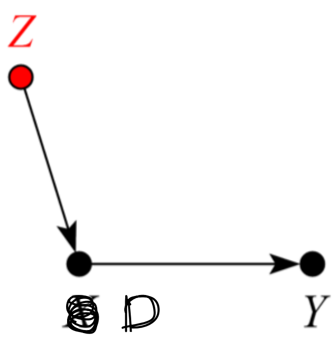
(3)

Control Neutral (Bueno para precisión)



(8)

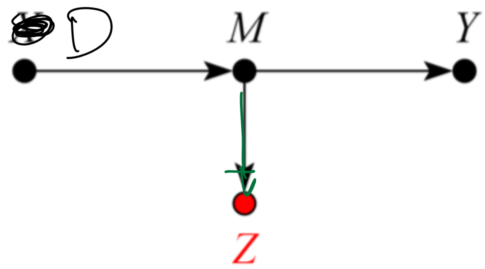
Control Neutral (malo para precisión)



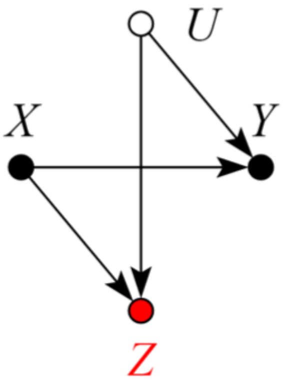
(9)

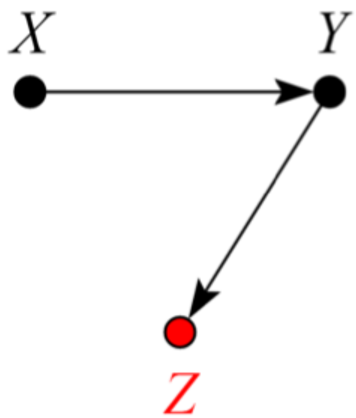
Malos Controles.

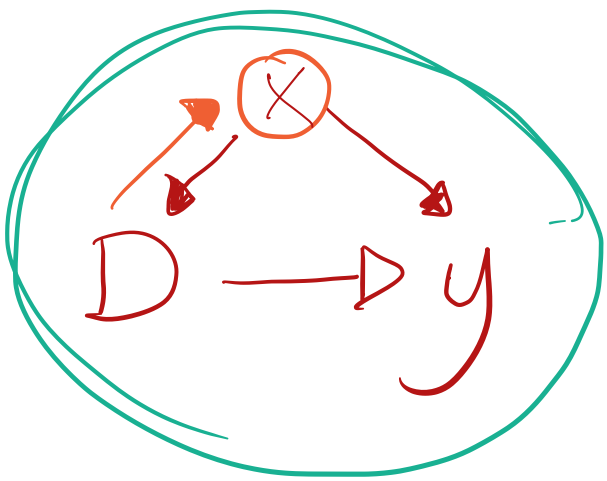
(11)



(12)







¿Qué variables incluir?

$X=$ Educación, Experiencia, Habilidad

Depende de la disponibilidad de los datos

Definiciones clave: ¿Qué es un buen control?

Buen Control Todas las características que NO se ven afectadas directamente por el tratamiento. Pre-determinadas

Mal control Variables afectadas por el tratamiento

Implicación de usar o NO un buen control $\Rightarrow$ una comparación de la var de resultado condicional en X NO tiene interpretación causal

Resultados Potenciales.

$$Y_i = \begin{cases} Y_i(D_i=1) & \text{Si participa} \\ Y_i(D_i=0) & \text{Si NO participa} \end{cases} \quad Y_i = D_i Y_i(D_i=1) + (1-D_i) Y_i(D_i=0)$$

$$Y(D_i=1), Y(D_i=0) \perp D$$

$$F_i: \begin{cases} F_i(D_i=1) & \text{Si participa} \\ F_i(D_i=0) & \text{Si NO participa.} \end{cases} \quad F_i = D_i F_i(D_i=1) + (1-D_i) \cdot F_i(D_i=0)$$

$$F(D_i=1), F(D_i=0) \perp D$$

$$E[Y_i \mid D=1] - E[Y_i \mid D=0] = E[Y(D=1)] - E[Y(D=0)] = E[Y(1)-Y(0)] = ATE$$

$$= E[Y(D=1) - Y(D=0)] = ATE$$

$$E[F_i \mid D=1] - E[F_i \mid D=0] = E[F(D=1)] - E[F(D=0)] = E[F(1)-F(0)] = ATE$$

$$= E[F(D=1)] - E[F(D=0)] = ATE$$

$$Y(D_i=0),\ Y(D_i=1),\ F(D_i=1),\ F(D_i=0) \perp D$$

Supuestos

- $y_i(D_i=1),\ y_i(D_i=0) \perp D_i$ //
- $F_i(D_i=1),\ F_i(D_i=0) \perp D_i$ //

$$y_i = \alpha + \tau D_i + \varepsilon$$

$$\hat{\tau}_{MCO} = E[y_i \mid D_i=1] - E[y_i \mid D_i=0]$$

$$\underbrace{\bar{y}_{D_i=1} - \bar{y}_{D_i=0}}$$

$$ATE = ATT = ATC.$$

$\rightarrow E[\hat{\tau}_{MCO}] = \tau$ insesgado

$\text{plim}[\hat{\tau}_{MCO}] = \tau$ consistente.

$$Var[\hat{\tau}] = \sigma_\varepsilon^2 (D'D)^{-1}$$

¿Qué pasa si incluimos a F como un control?

$$y = \alpha + \tau D + \beta F + \varepsilon$$

$$E[y \mid D=1, F=1] - E[y \mid D=0, F=1] = ?$$

$$y_i = \begin{cases} y_i\,(D_i=1) \\ y_i\,(D_i=0) \end{cases} \rightarrow y_i = D_i\, y(D_i=1) + (1-D_i)\,y(D_i=0)$$

$$F_i = \begin{cases} F_i\ (D_i=1) \\ F_i\ (D_i=0) \end{cases} \rightarrow F_i = D_i\, F_i(D_i=1) + (1-D_i)\,F_i\,(D_i=0)$$

$$E\left[D_i\, y(D_i=1) + (1-D_i)\, y_i(D_i=0) \mid D_i=1,\ [D_i F(D_i=1) + (1-D_i)\, F_i\,(D_i=0)] = 1\right.$$

$$-$$

$$E\left[D_i\, y(D_i=1) + (1-D_i)\, y_i(D_i=0) \mid D_i=0,\ [D_i F(D_i=1) + (1-D_i) F_i(D_i=0)] = 1\right.$$

$$= E\left[y(D_i=1) \mid \overset{①}{F(D_i=1)} = 1\right] -$$

$$E\left[y(D_i=0) \mid \overset{②}{F(D_i=0)} = 1\right] +$$

$$E\left[y(D_i=0) \mid \overset{③}{F(D_i=1)} = 1\right] -$$

$$E\left[y(D_i=0) \mid \overset{④}{F(D_i=1)} = 1\right]$$

$$= \underbrace{E\left[y(D_i=1) - y(D_i=0) \mid F(D_i=1)=1\right]}_{\text{ATE}} +$$

$$\underbrace{E\left[y(D_i=0) \mid F(D_i=1)=1\right] - E\left[y(D_i=0) \mid F(D_i=0)=1\right]}_{\text{Sesgo de Agrupación}}$$

Moraleja → $y = \alpha + \tau D + \underset{t-1}{X\beta} + \varepsilon$

$D \rightarrow$ STEM

$D \rightarrow$ NO STEM

$Y_S = \alpha + \tau_S D + \varepsilon$

$y_{NS} = \alpha + \tau_{NS} D + \varepsilon$

Malos controles

$$\begin{tikzcd} & Y \\ D \arrow[ur] \arrow[dr] & \\ & F \end{tikzcd}$$

$$Y(D=1),\ Y(D=0),\ F(D=1),\ F(D=0) \perp D$$

$$Y = \begin{cases} Y(D=1) & \text{si } D=1 \\ Y(D=0) & \text{si } D=0 \end{cases} \rightarrow Y = D\,Y(D=1) + (1-D)\,Y(D=0)$$

$$F = \begin{cases} F(D=1) & \text{si } D=1 \\ F(D=0) & \text{si } D=0 \end{cases} \rightarrow F = D\,F(D=1) + (1-D)\,F(D=0)$$

$$\underbrace{E[Y \mid D=1] - E[Y \mid D=0]}_{\text{Naive}} = E[Y(D=1) \mid D=1] - E[Y(D=0) \mid D=0]$$

$$= E[Y(D=1)] - E[Y(D=0)]$$

$$= \underbrace{E[Y(D=1) - Y(D=0)]}_{\text{ATE}}$$

$$= \underbrace{E[Y(D=1) - Y(D=0) \mid D=1]}_{\text{ATT}}$$

$$= E[Y(D=1) - Y(D=0) \mid D=0]$$

ATU

$$Naive = ATE = ATT = ATU$$

$$E[F \mid D=1] - E[F \mid D=0] = Naive =$$
$$= ATE = ATT = ATU$$

$$Y = \alpha + D\tau + \varepsilon$$

$$y = \alpha + D\tau + F\beta + u$$

$$E[Y \mid D=1, F=1] = E[D \cdot Y(D=1) + (1-D)Y(D=0) \mid D=1, DF(D=1) + (1-D)F(D=0) = 1]$$

$$= E[Y(D=1) \mid D=1, F(D=1)=1]$$
$$= E[Y(D=1) \mid F(D=1)=1]$$

$$E[Y \mid D=0, F=1] = E[DY(D=1) + (1-D)Y(D=0) \mid D=0, DF(D=1) + (1-D)F(D=0) = 1]$$

$$= E[Y(D=0) \mid D=0, F(D=0)=1]$$
$$= E[Y(D=0) \mid F(D=0)=1]$$

$$Y = \alpha + D\tau + F\beta + \varepsilon$$

$$E[Y \mid D=1, F=1] - E[Y \mid D=0, F=0]$$

$$= ①\ E[Y(D=1) \mid F(D=1)=1] \ -$$

$$②\ E[Y(D=0) \mid F(D=0)=1] \ +$$

$$③\ E[Y(D=0) \mid F(D=1)=1] \ -$$

$$④\ E[Y(D=0) \mid F(D=1)=1]$$

$$= \underset{(1-4)}{E[Y(D=1) - Y(D=0) \mid F(D=1)=1]} \ +$$

$$\underbrace{E[Y(D=0) \mid F(D=1)=1] - E[Y(D=0) \mid F(D=0)=1]}_{\text{Sesgo de agrupación}}$$

3 2

$$y = \alpha + \tau D + \beta F + \varepsilon$$ Ni de Riesgos

$$y = \alpha + \tau D + \varepsilon \qquad Cov(D, \varepsilon) \neq 0$$

$$E[y \mid D=1] = \alpha + \tau$$
$$E[y \mid D=0] = \alpha$$

$$ATE = E[y \mid D=1] - E[y \mid D=0]$$
$$= E[y(D=1) \mid D=1] - E[y(D=0) \mid D=0]$$
$$= \tau$$

$$E[\hat{\tau}] = \tau$$ insesgado

$$\operatorname{plim}[\hat{\tau}] = \tau$$ consistente.

$$Y = \alpha + \tau D + \beta F + \varepsilon$$

$$E[y \mid D=1, F=1] = E[D \cdot y(D=1) + (1-D)\, y(D=0) \mid D=1, [D \cdot F(D=1) + (1-D)$$

$$F(D=0)] = 1]$$

$$= E[y(D=1) \mid F(D=1)=1]$$

$$E[y \mid D=0, F=1] = E[D \cdot y(D=1) + (1-D) \cdot y(D=0) \mid D=0, [D \cdot F(D=1) + (1-D) F(D=0)] = 1]$$

$$= E[y(D=0) \mid F(D=0)=1]$$

$$E[y \mid D=1, F=1] - E[y \mid D=0, F=1]$$

1 $E[y(D=1) \mid F(D=1)=1]$ −

2 $E[y(D=0) \mid F(D=0)=1]$ +

3 $E[y(D=0) \mid F(D=1)=1]$

4 $-E[y(D=0) \mid F(D=1)=1]$

$$\underbrace{E[y(D=1)\mid F(D=1)=1] - E[y(D=0)\mid F(D=1)=1]}_{ATE_{\mid F=1}}$$

$$+\underbrace{E[y(D=0)\mid F(D=1)=1] - E[y(D=0)\mid F(D=0)=1]}_{\text{Sesgo de agrupación}}$$